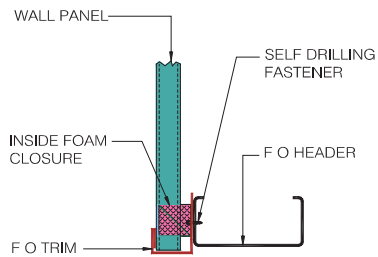


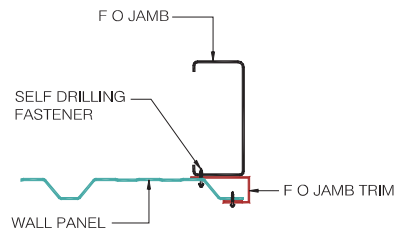
CHAPTER 13 : ACCESSORIES

SECTION 13.6 : Framed Openings

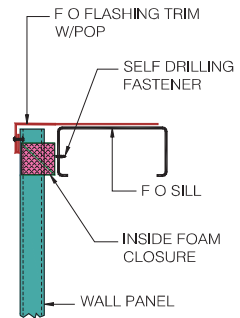
2 of 3



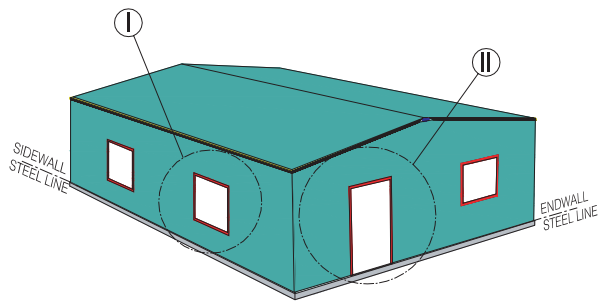
SECTION-A



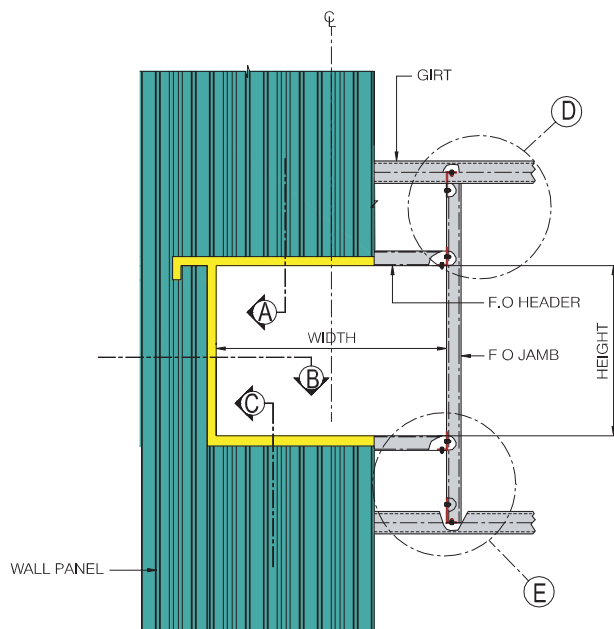
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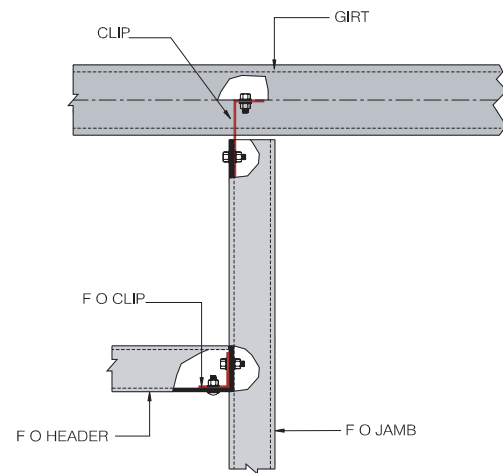
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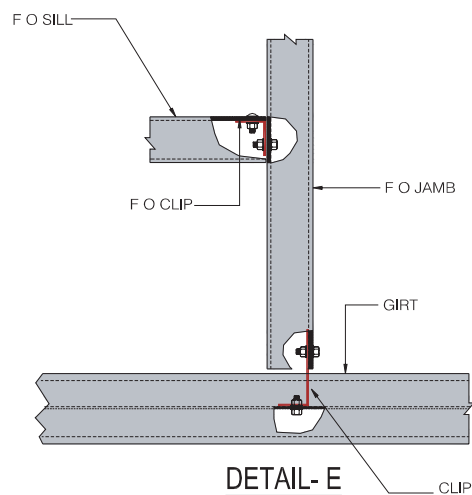
FRAMED OPENING IN WALL



DETAIL-I

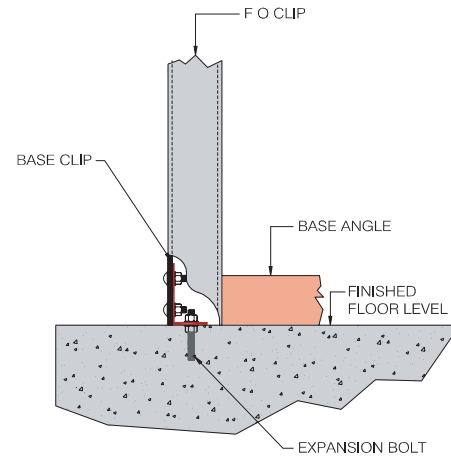
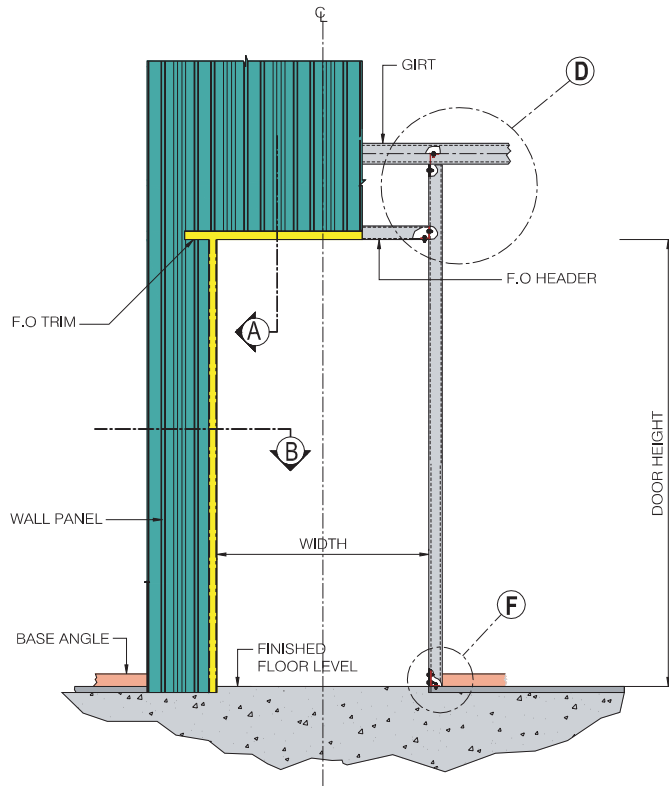


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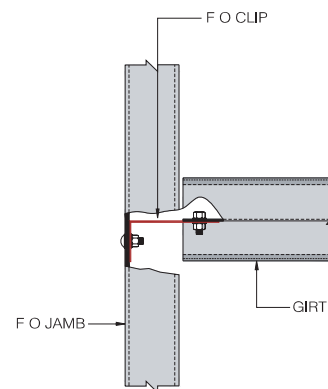
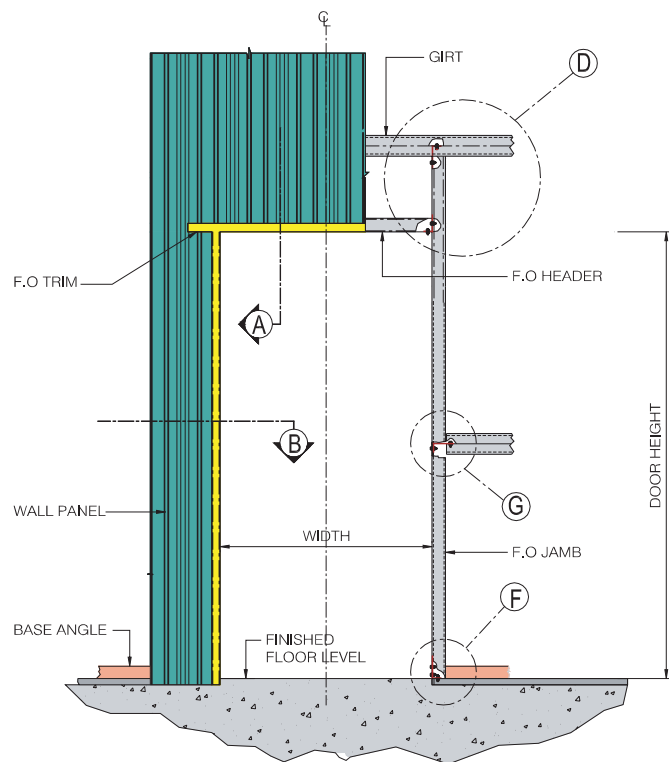
DETAIL-E

FRAMED OPENING IN WALL BETWEEN GIRTS



DETAIL-F

FRAMED OPENING IN WALL FROM FLOOR LINE TO FIRST GIRT LEVEL



DETAIL-G

FRAMED OPENING IN WALL FROM FLOOR LINE TO ANY HEIGHT

Ventilators: The primary purpose of ventilation is the control of the interior environment of the building by the removal / reduction of following:

- **Heat buildup**, thus providing comfort for workers, preserving goods and enabling equipment to function properly.
- **Gaseous by-products**, (the result of some manufacturing processes) thus providing a healthier atmosphere for workers.
- **Flammable fumes**, thus minimizing fire hazards (ventilation is also important after a fire has started at which time it helps in removing the fumes and smoke), and improving visibility for escapees and fire fighters.

Ventilation, whether natural or forced, does not have heat reduction characteristics. Air conditioning must be considered if a cooler temperature is desired.

Ventilation equipment comes in two categories:

- Inlet equipment
- outlet equipment

Either one can be natural or forced (power).

The planning and correct distribution of ventilation equipment has a major role in the overall efficiency of the ventilation system. Ventilation efficiency is also affected by the location of equipment, partitions and doors inside a building.

A detailed study of ventilation must be made at the planning stage of the building. For complex buildings, determining ventilation requirements may require vast calculations. Much simpler procedures normally suffice for typical metal buildings.

There are two simple and practical methods for determining ventilation requirements:

- **The Air Change Method**, which is based on a recommended rate of air changes per hour for different building usages when ventilator capacity is given.
- **The Heat Removal Method**, which calculates the volume of air required to remove the heat gain inside a building.

These methods are good tools to approximate the ventilation requirements of the building. More detailed and accurate ventilation plans should be sought by contracting ventilation specialists directly.

Table 1			
Stack Height (m)	Temperature Difference (°C)	R = Exhaust Capacity (m ³ /s)	
		MRV 300	MRV 600
3	5	0.779	1.559
6	5	0.916	1.832
	10	1.109	2.218
9	5	1.021	2.041
	10	1.257	2.514
	15	1.438	2.876
12	10	1.382	2.763
	15	1.591	3.182
15	10	1.492	2.983
	15	1.726	3.452

Ventilation Design Using Air Change Method

$$Q_v = \frac{V \times N}{R \times 3600} \text{ where,}$$

Q_v = Quantity of ventilators.

R = Exhaust capacity (m^3/sec).

V = Building volume (m^3).

N = Air change per hour.

In Table 1, enter the stack height, which is the average of the eave height and the ridge (peak) height. Enter the temperature difference between the inside and outside of the building. Read the equivalent required total exhaust capacity in m^3/sec .

The number of recommended air changes per hour is obtained from Table 2.

Finally, apply the aforementioned equation to determine the number of required ventilators. For efficient functioning of the ventilation system the free inlet area (permanent openings plus the effective area of the louvers) must be greater than 150% of the ventilation area.

Table 2		
Type of Building	N = Recommended Air Changes Per Hour	
	From	To
Warehouses, factories, dining rooms, machine shops, engine rooms, textile mills, wood working shops	5	10
Boiler rooms, paint shops, garages, schools	10	15
Buildings with fumes, kitchens, paper mills, textile mills, dye houses	15	20



Ventilation Design Using Heat Removal Method

Calculate the total heat gain (H) in the building:

$$H = H_s + H_l + H_i \quad \text{where,}$$

H_l = Heat gain from lighting

H_i = Heat gain from internal equipment, people, etc.

H_s = Solar Heat Gain

$$H_s = A \times I \times a \times e \quad \text{where,}$$

A = Roof Area

I = Actual Solar Radiation striking the roof surface (=1.12 kW/m² for UAE)

a = Solar radiation factor of the sheets (=0.40 for Mammut standard roof sheeting)

e = Proportion of "a" transmitted into the building. This factor is dependent on the "U" value of insulation

= 0.093 for a building with insulation

= 1.000 for a building without insulation

Calculate the ventilation rate required ($VR_{req'd}$):

$$VR_{req'd} = \frac{H}{S_H \times D \times T} \quad \text{where,}$$

S_H = Specific heat capacity of air
= 1.005

D = Density of air = 1.206 kg/m³

T = Temperature difference (use °C difference per meter of stack height).

Determine the quantity of ventilators needed,

Quantity of ventilators

$$= \frac{VR_{req'd}}{R}$$

where,

R = Exhaust capacity in m³/s of all ventilators (see page 2 of 13 of this **section**)

Calculate the total area of gravity ventilators:

Total ventilator area

$$= \frac{VR_{req'd}}{[0.03 \times \text{stack height} \times T]^{1/2}}$$

For special ventilators, the supplier's catalogue must be consulted.



Example of Ventilation Design using the Air Change Method

A Clear Span building that is used as a factory requires ventilation using gravity ridge ventilators. The building is 140 m long and 36 m wide. The eave height is 9 m. The roof has a 0.5/10 slope.

The temperature difference between the inside and outside of the building is not expected to exceed 10°C. Determine the number and size of gravity ventilators that are needed.

The average of the eave height and the ridge height.	Stack Height	= 9.45 m
Exhaust capacity (R) at 9.45 m stack height is determined by interpolation from the table on page 1 of 13 of this section.	R (for MRV 300) R (for MRV 600)	= 1.276 m ³ /s = 2.551 m ³ /s
The recommended number of air changes per hour (N) for factories is 5 to 10 times.	N	= 8
Volume of building.	V	= 140 x 36 x 9.45 = 47628 m ³
Quantity of MRV-300 ventilators required. Need 83 MRV-300 ventilators	Q _v	= $\frac{(47628 \times 8)}{1.276 \times 3600}$ = 82.32 ~ 83
The maximum number of ridge ventilators (Q_{max}) that the building can accommodate. Q_{max} = $\frac{\text{Building Length (m)} - 1}{\text{Ventilator Length (m)}}$	Q _{max}	= $\frac{140-1}{3}$ = 46.33 ~ 46
MRV-300 Cannot be used. Try MRV-600	Q _{max} < Q _v	= 46 < 83
Quantity of MRV-600 ventilators required. Q_v < Q_{max} = 46	Q _v	= $\frac{(47628 \times 8)}{2.551 \times 3600}$ = 41.49 ~ 42
Use 42 gravity ridge ventilators MRV-600 for the building		